individual assignment 5

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Set up

Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")
```

```r
# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning

`sites` object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

`taxon_list` object

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

`water_quality` object

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
```

```
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

`aq_ins` object

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
```

```
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

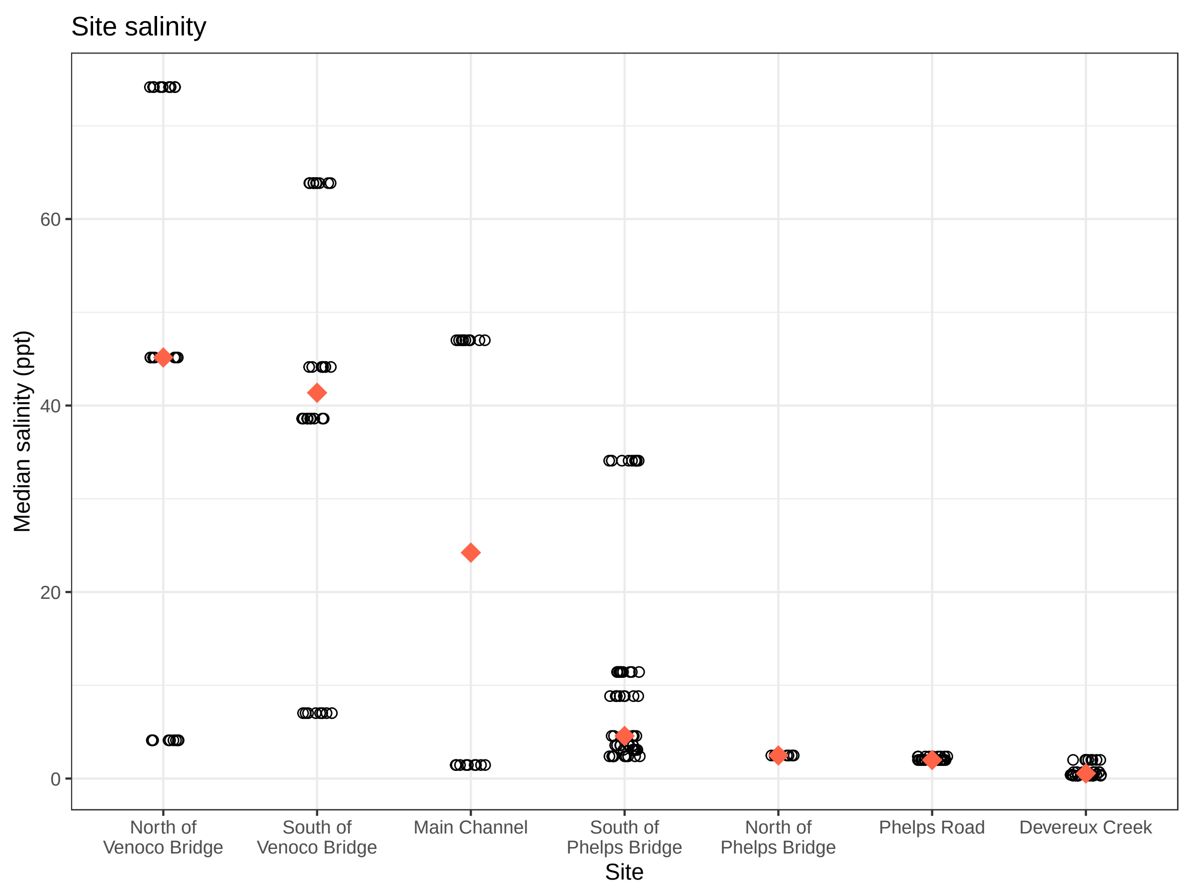
Inserting the needed class code

Salinity visual

```
# base layer
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1,
              size = 2) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent median
  stat_summary(fun = median,
               color = "tomato",
```

```
                 size = 1,
                 shape = 18) + # 18 = filled diamond
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  # different theme
  theme_bw()

# displaying object
salinity
```



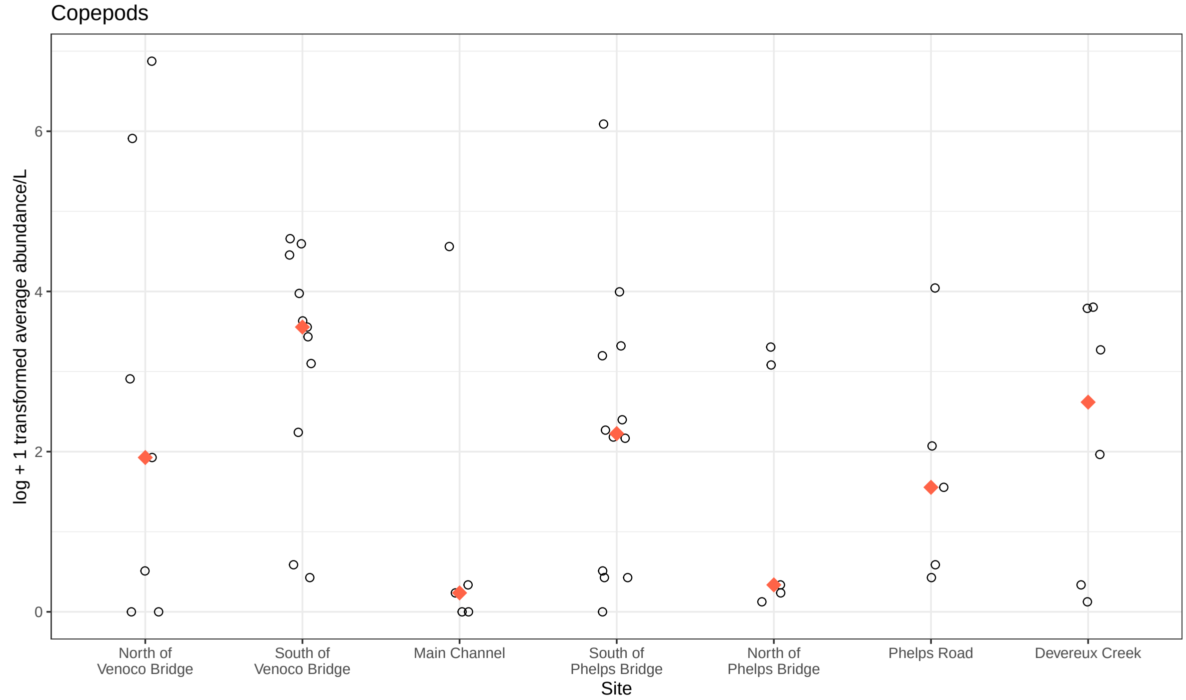
Copepod cleaning

```r
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

Copepod visual

```r
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1,
              size = 2) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "tomato",
               shape = 18) + # 18 = filled diamond
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # different theme
  theme_bw()

# displaying object
copepods
```



Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site
- y-axis: average abundance/L
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
# creating new object from clean aquatic inverts
ostracod <- aq_ins_clean |>
  # filter to only include ostracods
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

```r
# display 10 random rows
slice_sample(ostracod, n = 10)
```

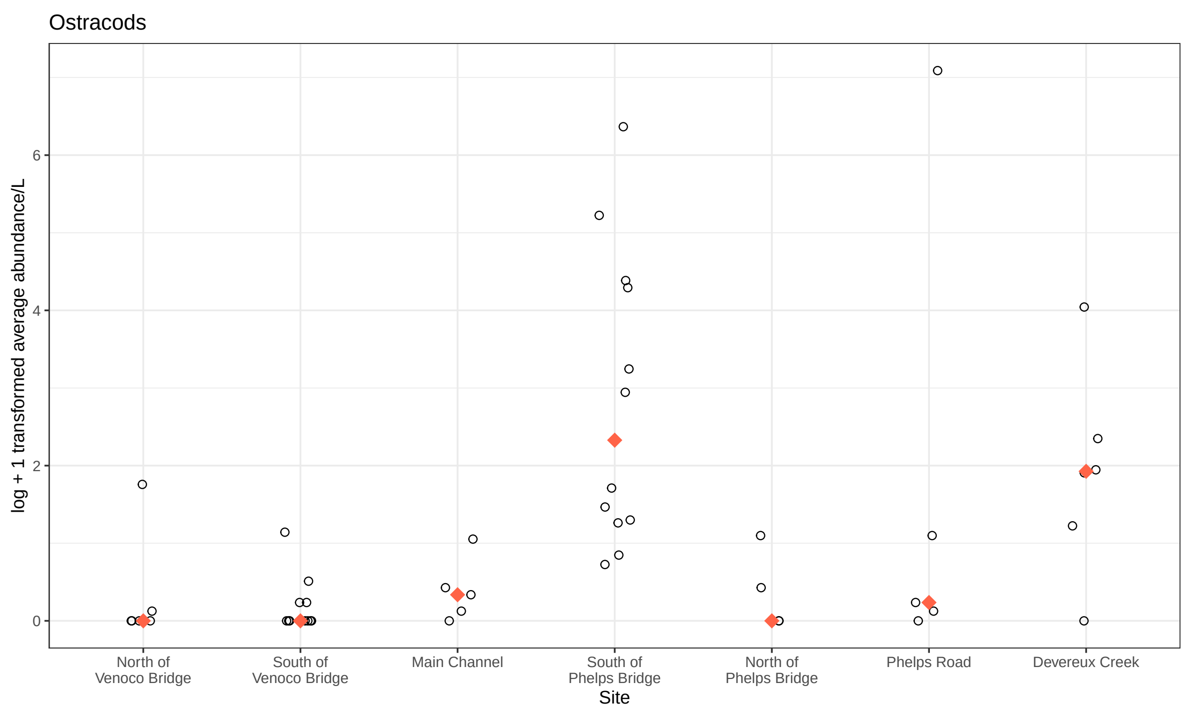
```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-02-25_NEC  2022-02-25 00:00:00 NEC   ostr~   0       8.75  44.1     9.87
 2 2023-11-12_NMC  2023-11-12 00:00:00 NMC   ostr~   0.533  NA     47      14.0 
 3 2022-05-11_NDC  2022-05-11 00:00:00 NDC   ostr~   0       9.11   0.398   9.4 
 4 2021-04-15_NPB  2021-04-15 00:00:00 NPB   ostr~ 185.      8.09   3.57    8.24
 5 2021-09-01_NPB2 2021-09-01 00:00:00 NPB2  ostr~   0      NA     NA      NA   
 6 2022-05-13_NVBR 2022-05-13 00:00:00 NVBR  ostr~   0       9.72  74.2    11.4 
 7 2022-01-21_NEC  2022-01-21 00:00:00 NEC   ostr~   0      NA     NA      NA   
 8 2023-05-04_NMC  2023-05-04 00:00:00 NMC   ostr~   1.87    8.32  NA      10.3 
 9 2022-02-21_NPB  2022-02-21 00:00:00 NPB   ostr~   1.33    7.39  34.1     0.34
10 2022-08-10_NPB  2022-08-10 00:00:00 NPB   ostr~   2.67    8.28   8.84    9.81
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```r
# creating the ostracod figure
ostracod_plot <- ggplot(data = ostracod,
                        # x-axis: site
                        # # y-axis: mean abundance/L
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
  # first layer: individual observations as open circles
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1,
              size = 2) +
  # second layer: median as a filled blue diamond
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "tomato",
```

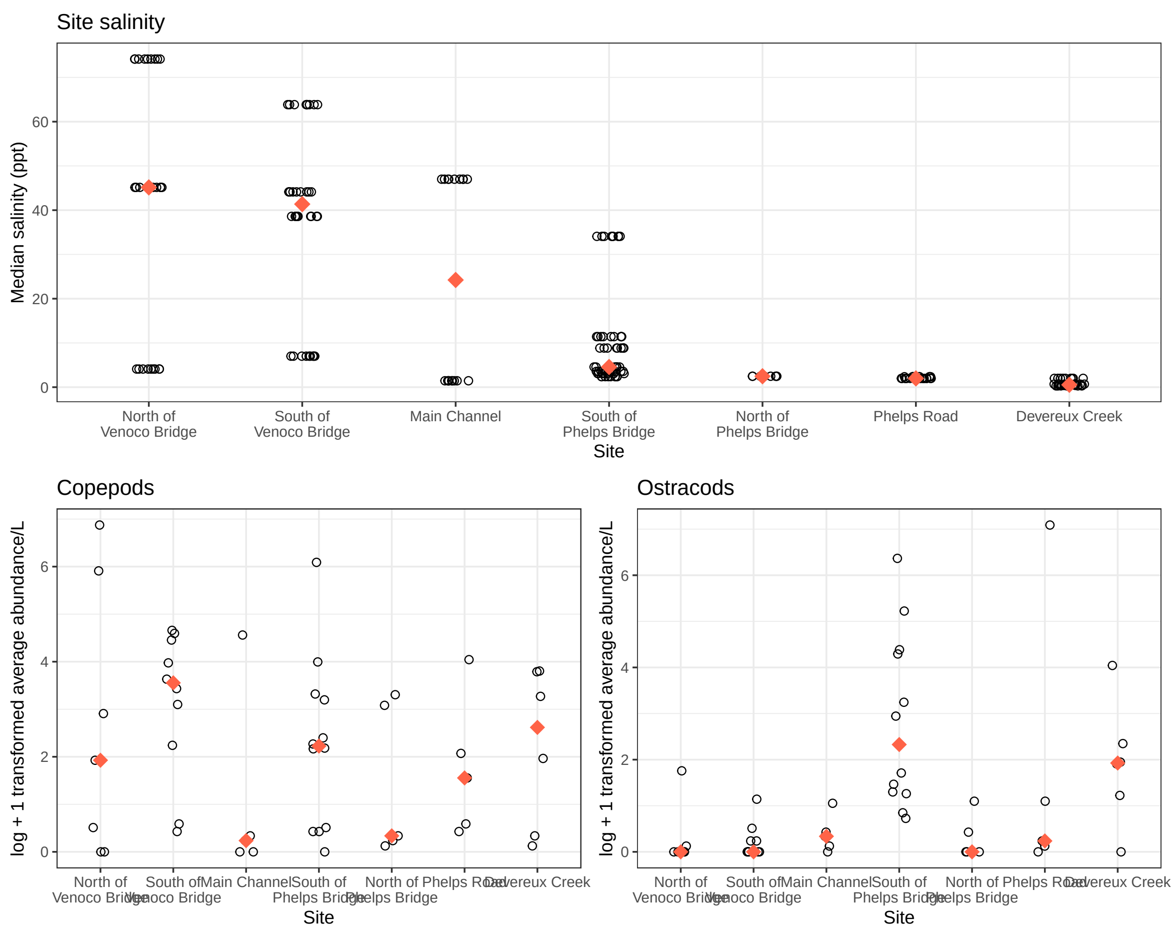
```
                shape = 18) + # 18 = filled diamond
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # axis labels and title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # different theme
  theme_bw()

# displaying object
ostracod_plot
```



d. Create a multipanel figure

```
# combining the figures together
salinity/ ( copepods + ostracod_plot )
```



e. Interpret your figure

- Differences between sites in copepod and ostracod abundance:

Copepod abundance was highest at South of Venoco Bridge with a median above 3, while ostracod abundance peaked at South of Phelps Bridge with a median above 2. Overall, ostracods had lower median abundance than copepods at every site, suggesting copepods are the more dominant group across this system.

- How ostracod abundance may relate to salinity:

Ostracod abundance does not show a clear relationship with salinity — while most sites had low ostracod abundance regardless of their salinity level, South of Phelps Bridge had the highest ostracod abundance despite having low salinity (below 7 ppt), similar to several other low-salinity sites where ostracods remained scarce. This suggests that salinity alone does not explain the variation in ostracod abundance across sites, and other environmental factors may be at play.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: salinity
- y-axis: log + 1 transformed average abundance/L
- color: taxon
- geometry: `geom_point()`
- facets: `facet_wrap(~ taxon, scales = "free")`

b. Wrangle your data

c. Code your figure

d. Interpret your figure